

Practical No. 1(Install any one embedded system(ex- Arduino IDE)and execute program to turn LED ON/OFF using delay) :

```
int led = 8;

void setup() {
  pinMode (led, OUTPUT);
}

void loop() {
  digitalWrite(led, HIGH);
  delay (1000);
  digitalWrite(led, LOW);
  delay(1000);
}
```

Practical No. 2(Change the color of LED) :

```
int redPin= 11;
int greenPin = 10;
int bluePin = 9;

void setup() {
  pinMode(redPin, OUTPUT);
  pinMode(greenPin, OUTPUT);
  pinMode(bluePin, OUTPUT);}

void loop()
{
  setColor(255, 0, 0);
  // Red Color delay(1000);
  setColor(0, 255, 0);
  // Green Color delay(1000);
  setColor(0, 0, 255);
  // Blue Color delay(1000);
  setColor(170, 0, 255);
}
```

```
// Purple Color delay(1000);
}

void setColor(int redValue, int greenValue, int blueValue)
{
  analogWrite(redPin, redValue);
  analogWrite(greenPin, greenValue);
  analogWrite(bluePin, blueValue);
}
```

Practical No. 3(Control the
brightness of LED using
PWM Techniques):

```
#define led 11 void
setup(){ pinMode(led,
OUTPUT);} void loop(){
for(int i = 0; i <= 255; i++)
{ analogWrite(led, i);
delay(15);}}
```

Practical No. 4:
(Detect the
presence or
absence of Light
using LDR Sensor
)

```
int ldrPin = A0; int
led = 7; int
threshold = 70;
```

```

void setup(){
  Serial.begin(9600);
  pinMode(led, OUTPUT);
} void loop(){ int data =
  analogRead(ldrPin);
  Serial.println("");
  Serial.print("Light Sensor ");
  Serial.print("Value = ");
  Serial.print(data); if(data
  <= threshold){
  digitalWrite(led, HIGH); }
  else{ digitalWrite(led,
  LOW);}}

```

Practical No. 5

(Measure the
temperature of
the object)

```

float temp; int
tempPin = A1;
void setup() {
  Serial.begin(9600); } void
loop() { temp =
  analogRead(tempPin); temp =
  temp * 0.48828125;
  Serial.print("TEMPERATURE = ");
  Serial.print(temp);

```

```
Serial.print("*C");  
Serial.println();  
delay(1000);  
}
```

Practical No. 6

(Sense the touch of finger when it
is placed on board)

```
#define ctsPin 2 int ledPin = 13;  
void setup() { Serial.begin(9600);  
pinMode(ledPin, OUTPUT);  
pinMode(ctsPin, INPUT);} void  
loop() { int ctsValue =  
digitalRead(ctsPin); if (ctsValue  
== HIGH) { digitalWrite(ledPin,  
HIGH);  
Serial.println("TOUCHED");}  
else{ digitalWrite(ledPin,LOW);  
Serial.println("not touched");}  
delay(500);}
```

Practical No. 7

(Detect the obstacle using IR sensor)

```
int IRSensor = 7; void  
setup(){  
pinMode (IRSensor, INPUT);  
Serial.begin(9600);} void loop(){ int  
statusSensor = digitalRead (IRSensor); if  
(statusSensor == 1){ Serial.println("obstacle
```

```
detected");} else{Serial.println(" No obstacle  
detected");}}
```

Practical No. 8

(Measure the Distance between sensor
and object using ultrasonic sensor)

```
#define echoPin 12  
  
#define trigPin 13 long duration;  
int distance;  
void setup()  
{ pinMode(trigPin, OUTPUT);  
  pinMode(echoPin, INPUT);  
  Serial.begin(9600);  
  Serial.println("Distance measurement using Arduino Uno.");  
  delay(500);}  
  
void loop(){  
  digitalWrite(trigPin, LOW);  
  delayMicroseconds(2);  
  digitalWrite(trigPin, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(trigPin, LOW);  
  duration = pulseIn(echoPin, HIGH);  
  distance = duration * 0.0344 / 2;  
  Serial.print("Distance: ");  
  Serial.print(distance);  
  Serial.println(" cm");  
  delay(100);}
```

Practical No. 9:

(Detect the
presence of Gas)

```
int LED = 11;
const int gas = 0;
int MQ2pin = A0;
void setup() {
  Serial.begin(9600)
;} void loop() {
  float
  sensorValue,MQ2
  pin; sensorValue =
  analogRead(MQ2
  pin);
  if(sensorValue >=
  470){
    digitalWrite(LED,
    HIGH);
    Serial.println(" |SMOKE DETECTED");}
  else{ digitalWrite(LED,LOW);
    Serial.println("Sensor Value: ");
    Serial.println(sensorValue);}
  delay(1000);} float getsensorValue(int
  pin){ return (analogRead(pin));}
```

Practical No. 11:

(Change the status of Buzzer ON/OFF)

```
#define BUZZER 2
#define DELAYTIME 1000
void setup() {
```

```
pinMode(2, OUTPUT);  
digitalWrite(2, LOW);}   
void loop(){ digitalWrite(2,  
HIGH);  
delay(DELAYTIME);  
digitalWrite(2, LOW);  
delay(DELAYTIME);}
```

Practical No. 13

(Display the
message as per
detection of
motion of object)

```
int sensorState =  
0; void setup(){  
pinMode(2,  
INPUT);  
Serial.begin(9600);  
} void loop(){  
sensorState =  
digitalRead(2); if  
(sensorState ==  
HIGH){  
Serial.println("motion detected");  
} else{  
Serial.println("No motion detected");  
} delay(10);  
}
```

Practical No. 14

(Control relay state based on input from IR Sensor):

```
int IRSensor = 7; int relay = 2; void
setup(){ pinMode (IRSensor, INPUT);
pinMode (relay, OUTPUT);
Serial.begin(9600);} void loop(){ int
statusSensor = digitalRead (IRSensor); if
(statusSensor == 1){
digitalWrite(relay,HIGH);
} else{
digitalWrite(relay,LOW);}}
```

Practical No. 15

(Switch the LED ON/OFF on detection of obstacles using PIR sensor):

```
int PIRSensor = 2;
int LED = 13;
void setup(){
pinMode(LED, OUTPUT);
pinMode(PIRSensor, INPUT);
Serial.begin(9600);}
void loop(){
digitalWrite(2, LOW);
Serial.println(digitalRead(2));
if (digitalRead(2) == 1) {
    digitalWrite(13, HIGH);
}
else {
```



```
digitalWrite(13, LOW);} delay(10);}
```

Practical No. 16(Measure the Distance between sensor and object and sound buzzer when obstacle is detected in some specified range of distance) :

```
const int trigPin = 3;    const int echoPin = 2;    const int buzzerPin =  
7; long duration;    int distance;  
int minDistance = 10; // Minimum distance for obstacle detection (in centimeters)  
int maxDistance = 50; // Maximum distance for obstacle detection (in  
centimeters) void setup() {    pinMode(trigPin, OUTPUT);  
pinMode(echoPin, INPUT);    pinMode(buzzerPin, OUTPUT);  
} void loop()  
{  
digitalWrite(trigPin, LOW);    delayMicroseconds(2);  
digitalWrite(trigPin, HIGH);    delayMicroseconds(10);  
digitalWrite(trigPin, LOW); duration = pulseIn(echoPin,  
HIGH); distance = duration * 0.034 / 2; if (distance >=  
minDistance && distance <= maxDistance) {  
digitalWrite(buzzerPin, HIGH);  
} else {  
digitalWrite(buzzerPin, LOW);}  
delay(100);}
```

Practical No.17

(Activate the alarm

system if smoke

detected)

: int val = A0; void

setup(){ pinMode

(2,OUTPUT);

pinMode (A0,INPUT);

```

Serial.begin(9600);
}void loop(){ val =
analogRead (A0); if
(val <500 ){
digitalWrite (2,
LOW);} else{
digitalWrite(2,
HIGH);}

```

Practical No. 19

(Upon detection of
fire, activate the
LED indicator and
initiate the alarm
system)

```

: int sensorValue =
0; void setup() {
pinMode(8, OUTPUT); pinMode(9,
OUTPUT);
Serial.begin(9600); } void loop() {
int sensorValue = analogRead(A0);
Serial.println(sensorValue, DEC);
if (sensorValue > 250) {
digitalWrite(8, HIGH);
digitalWrite(9, HIGH);
} else { digitalWrite(9,
LOW); digitalWrite(8,
LOW);}}

```